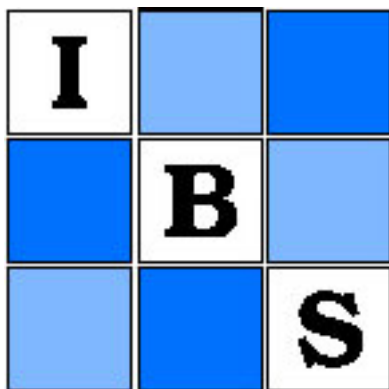




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A Random Walk Rule for Phase I Clinical Trials

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SUMMARY

We describe a family of random walk rules for the sequential allocation of dose levels to patients in a dose–response study, or phase I clinical trial. Patients are sequentially assigned the next higher, same, or next lower dose level according to some probability distribution, which may be determined by ethical considerations as well as the patient’s response. It is shown that one can choose these probabilities in order to center dose level assignments unimodally around any target quantile of interest. Estimation of the quantile is discussed; the maximum likelihood estimator and its variance are derived under a two-parameter logistic distribution, and the maximum likelihood estimator is compared with other nonparametric estimators. Random walk rules have clear advantages: they are simple to implement, and finite and asymptotic distribution theory is completely worked out. For a specific random walk rule, we compute finite and asymptotic properties and give examples of its use in planning studies. Having the finite distribution theory available and tractable obviates the need for elaborate simulation studies to analyze the properties of the design. The small sample properties of our rule, as determined by exact theory, compare favorably to those of the continual reassessment method, determined by simulation.

1. Introduction

Pharmacological toxicity studies are designed to examine the relationship between the dose of a therapeutic agent and the potential toxic effect of that agent on some population. When the total number of subjects is fixed and a set of possible dose levels is given, an experimental design is completely determined by the rule that allocates dose levels to subjects. Such designs are among those called dose–response designs. A principal goal of many such studies is to estimate a quantile, the dose level at which a desired probability of response is attained.

Consider a dose–response design in which subjects are sequentially assigned to one of K dose levels of a drug and subject response is dichotomous (toxic/nontoxic). Durham and Flournoy (1994) describe a family of designs where subjects are sequentially assigned to the next higher, same, or next lower dose level, with some probability that is dependent upon the previous subject’s response. Their designs are a randomized extension of Dixon and Mood’s (1948) up-and-down design. They belong to the class of random walk rules because the allocation scheme induces a random walk on the lattice of dose levels. Particular random walk rules have been studied by Derman (1957) and Tsutakawa (1967, 1972), for example.

Strategies for sequential dose–response designs include up-and-down methods, stochastic approximation methods, maximum likelihood methods, and Bayesian methods. A brief history is as

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follows. The first up-and-down designs were proposed for explosives testing as early as 1946 by Anderson, McCarthy, and Tukey and for psychophysics in 1947 by Nobel laureate von Békésy. Motivated by the explosives testing problem, Dixon and Mood (1948) obtained some elegant results by focusing on estimators of the median. Tsutakawa (1967, 1972) analyzed a grouped version of this up-and-down design using random walk theory. Derman (1957) first demonstrated that dose levels could be centered around any given target quantile by randomizing the up-and-down designs using a biased coin. Robbins and Monro (1951) developed the stochastic approximation method, which in this context employs an iterative nonparametric procedure to converge to a target quantile. The idea of estimating the target after each trial and performing the next experiment at that point was proposed as early as 1968 by Hall. Although several variations of this proposal using maximum likelihood estimation have been examined in the psychophysics and acoustics literature, only recently have we seen it in the statistics literature (see Wu, 1985). Bayesian methods for selecting the lattice of possible dosages were developed by Tsutakawa (1980) and were later applied in bone marrow transplantation (see Flournoy, 1993). For a recent review of Bayesian designs, see Chaloner and Verdinelli (1995).

Recently, much attention in the biomedical community has been focused on the continual reassessment method (CRM) of O'Quigley, Pepe, and Fisher (1990), a Bayesian sequential design that is similar to the QUEST procedure of Watson and Pelli (1983). The literature is replete with simulation studies analyzing properties of the CRM (see, e.g., Chevret, 1993; Faries, 1994; Goodman, Zahurak, and Piantadosi, 1995; Korn et al., 1994; O'Quigley, 1992; O'Quigley and Chevret, 1991; as well as the original paper).

Random walk rules are particularly attractive for the design of dose-response studies for several reasons: (1) the designs are essentially nonparametric; (2) exact finite and asymptotic distribution theory is completely worked out, which will allow the experimenter to choose design parameters for the most ethical allocation scheme; (3) specific designs can be chosen that allow the chosen design points to be distributed unimodally around a quantile of interest; (4) the designs are very simple to implement; and (5) they operate on a finite lattice of dosages.

Most of the theory for random walk rules has already been developed in a sequence of recent papers in the applied probability and sequential design literature (Durham and Flournoy, 1993, 1994, 1995; Durham, Flournoy, and Montazer-Haghighi, 1993, 1995; Flournoy, Durham, and Rosenberger, 1995; Rosenberger, Flournoy, and Durham, 1997). The purpose of this paper is to illustrate the theoretical results for a specific random walk rule, a biased coin up-and-down rule, and show how it can be used to develop appropriate, ethically attractive designs for dose-response studies/phase I clinical trials.

This paper is organized as follows. In Section 2, we describe a random walk rule for phase I studies. In Section 3, we compare finite properties of the random walk rule with simulated properties of the continual reassessment method. In Section 4, we discuss quantile estimation. In Section 5, we use our methodology to redesign a study on bone marrow transplantation. Finally, in Section 6, we discuss some practical implications of our results, some concerns, and questions for future research.

2. A Random Walk Rule

Let $x_1, \dots, x_K$ be an ordered set of doses. As in most dose-response methodology, we assume that both efficacy and toxicity are monotonically related to dose. Hence, our goal is to find the highest tolerable (defined by investigators) dose. Durham and Flournoy (1994) describe a broad class of random walk rules, of which the Biased Coin Design II (BCD) (Durham and Flournoy, 1995, p. 145) is a special case and is as follows.

If patient j has just been assigned to dose level x_i , assign patient $j + 1$ with the following rule: If patient j experienced a toxic response, then assign patient $j + 1$ to level x_{i-1} . If patient j had no toxic response, then flip a biased coin with probability of heads $b \in [0, 0.5]$. If it lands heads up, assign patient $j + 1$ to level x_{i+1} . If it lands heads down, assign patient $j + 1$ to x_i .

It is necessary to impose boundary conditions, as one cannot lower the dose level at x_1 or raise the dose level at x_K . For our purposes, we simply assign the patient to x_1 or x_K , respectively, if such conditions occur.

This is a fairly conservative scheme in that one always lowers the dose level if there is a toxicity. For a nontoxicity, b is the probability of increasing the dose level. Shortly it will be shown how to set b in order to target a specific quantile of the dose-response curve.

Let p_i be the probability of toxicity at dose level x_i . Let $\mathbf{p} = \{p_1, \dots, p_K\}$. The distribution of $N_i(n)/n$, the proportion of subjects treated at dose level x_i , will be of great interest to us for ethical reasons, as we want to develop a rule that will place as few patients as possible at risk of

being assigned to highly toxic dose levels. Asymptotic properties of $N_i(n)/n$ are given in Durham and Flournoy (1995). In particular, the asymptotic treatment allocation proportions are given by $\pi = \{\pi_1, \dots, \pi_K\}$ in a closed-form function of $\mathbf{p}$. Of greater interest are finite distributional results (Durham et al., 1993, 1995), including the mean, variance, and covariances of $N_i(n)/n$ for any n . These are useful in determining the probability of assigning patients to dose levels in the subspace $\omega_x = \{x_\omega, x_{\omega+1}, \dots, x_K\}$, where x_ω is some dose level that is considered undesirable.

Another primary ethical parameter of interest is $R(n)/n$, the total proportion of toxic responses. Rosenberger and Sriram (1997) establish the almost sure convergence of $R(n)/n$ to $\mathbf{p} \cdot \pi$, while Flournoy et al. (1995) give the mean and variance of $R(n)$ for any n , together with a central limit theorem.

Thus, exact finite and asymptotic distributional properties are known that are useful in checking the ethical consequences of a particular design. We compute these via a program in MATLAB for any $\mathbf{p}$ and any number of design points.

Let μ be an unknown quantile of the dose-response curve corresponding to $\Gamma \in (0, 1)$; that is, μ is the maximum dose level deemed tolerable. A goal of the BCD is to center the frequency distribution of design points (i.e., treatments) around a specific, but unknown, quantile. For BCD, choosing b such that $b = \Gamma/(1 - \Gamma)$, $\Gamma \leq 0.5$, will asymptotically center the allocation proportions unimodally around μ , whenever μ is properly contained in the dose space (cf., Durham and Flournoy, 1994). For instance, if one wants to target the 33rd percentile of the dose-response curve, then by selecting $b = 1/2$, the asymptotic distribution of dose level assignments will be centered around μ .

3. Small Sample Comparison of BCD and CRM

In this section, we demonstrate the utility of having finite distribution theory for the operating characteristics of the BCD by comparing its results to results for the CRM obtainable only via extensive simulation. We compare expected proportions assigned at doses $x_1, \dots, x_6$ and expected proportion of toxicities for three different $\mathbf{p}$'s and $\Gamma = 0.20$. In each case, O'Quigley et al. (1990) ran 200 simulations to determine operating characteristics of the CRM. We compare the results found in their Table 3 (1990, p. 43) to the operating characteristics of the BCD using exact distribution theory for $n = 25$. The programs to compute finite moments take only seconds to run. The CRM simulations started each experiment at dose level x_3 .

We chose two starting rules for the BCD: (1) start at dose level x_3 , as did O'Quigley et al. (1990); and (2) treat the first subject at x_1 , then escalate dose levels for each subsequent subject until the first toxicity occurs; begin the BCD at that level, as did Korn et al. (1994) in their simulation of a modified CRM. Results are found in Table 1. O'Quigley et al. (1990) give no measures of variability in their simulation results. Exact variances of the allocation proportions and total toxicity can be easily found for the BCD using our software. We examined three sets of $\mathbf{p}$: for scenario A, $\mathbf{p} = (0.05, 0.10, 0.25, 0.35, 0.50, 0.70)$; for scenario B, $\mathbf{p} = (0.00, 0.00, 0.04, 0.09, 0.24, 0.49)$; and for scenario C, $\mathbf{p} = (0.30, 0.40, 0.52, 0.61, 0.76, 0.87)$.

When comparing starting rule (1) for the BCD with the same starting rule for the CRM, the similarities are compelling. Expected total toxicity differed by, at most, 3%, with the BCD doing slightly better under scenarios A and B. Expected allocation proportions were also very close. For the BCD with starting rule (2), the expected proportion of toxicities was less for scenario C, but slightly worse than starting rule (1) for scenarios A and B. Under scenario C, we have a case where the target quantile is out of range of the space of doses. Clearly, we cannot sample around the target quantile in this case using either the BCD or CRM. However, the mode is at the extreme end of the dose range closest to μ . For the other two cases, the unimodality property around μ for the BCD seems to hold well for these small sample studies.

4. Quantile Estimation

4.1 Maximum Likelihood

Rosenberger et al. (1997) show the consistency, efficiency, and asymptotic joint normality of maximum likelihood estimators $\hat{p}_i$ of p_i from a random walk rule. Assume the response probabilities are given by a location-scale family; that is, when

$$p_i = F_i \equiv F\left(\frac{x_i - \alpha}{\beta}\right), \quad (1)$$

for some distribution function F and constants α and β . For the location-scale family in (1), it is easy to see that the maximum likelihood estimator of μ is given by

$$\hat{\mu} = \hat{\beta}F^{-1}(\Gamma) + \hat{\alpha}, \quad (2)$$

Table 1
Biased coin design (BCD; exact expected values) versus continual reassessment method (CRM; simulated results from O’Quigley, 1990), under starting rules (1) and (2) for $n = 25$ and $\Gamma = 0.20$. Values reported in percentages.

i	$E\{N_i(n)/n\}$		
	CRM (1)	BCD (1)	BCD (2)
A. $p = [0.05, 0.10, 0.25, 0.35, 0.50, 0.70]$			
1	5.34	9.03	13.32
2	17.58	25.91	24.67
3	36.56	36.70	28.21
4	33.82	20.68	21.51
5	6.70	6.57	9.63
6	0.00	1.10	2.65
E{total toxicity}	22	21	22
B. $p = [0.00, 0.00, 0.04, 0.09, 0.24, 0.49]$			
1	0.50	0.00	4.00
2	1.02	3.90	5.83
3	9.06	26.69	13.64
4	35.08	35.53	30.90
5	52.46	25.22	30.93
6	1.88	8.65	14.70
E{total toxicity}	17	15	18
C. $p = [0.30, 0.40, 0.52, 0.61, 0.76, 0.87]$			
1	73.88	52.64	59.44
2	13.56	29.07	27.50
3	9.04	15.06	9.57
4	3.28	2.85	2.74
5	0.24	0.35	0.65
6	0.00	0.00	0.35
E{total toxicity}	34	37	36

where $\hat{\alpha}$ and $\hat{\beta}$ are the maximum likelihood estimators of α and β , respectively. Assuming that the maximum likelihood estimators exist, the asymptotic normality of $\hat{\mu}$ follows from the continuous mapping theorem (cf., Pollard, 1984, p. 46).

When the p_i ’s follow a logistic distribution, i.e.,

$$p_i = 1 - \left(1 + \exp \left\{ \frac{x_i - \alpha}{\beta} \right\} \right)^{-1}, \tag{3}$$

one can use a standard logistic regression package to find $\hat{\alpha}$ and $\hat{\beta}$ and compute $\hat{\mu}$ via (2) for any value of Γ . In the Appendix, we derive the usual approximation to the asymptotic variance from the Fisher’s information matrix, which suggests the following natural estimator of the variance of $\hat{\mu}$:

$$\widehat{\text{var}}(\hat{\mu}) = \frac{\hat{\beta}^2 \sum_{i=1}^K N_i(n) \hat{p}_i \hat{q}_i (x_i - \hat{\mu})^2}{\sum_{i < j} N_i(n) N_j(n) \hat{p}_i \hat{q}_i \hat{p}_j \hat{q}_j (x_i - x_j)^2}, \tag{4}$$

where $\hat{q}_i = 1 - \hat{p}_i$ and $\hat{p}_i$ is the observed proportion of toxicities at dose level x_i . Of course, expressions analogous to (4) can also be derived under a probit, Weibull, exponential, or other appropriate model.

4.2 Nonparametric Estimators

As random walk rules place design points unimodally around the target quantile, this suggests that the empirical mode of the assignment distribution would be one possible estimator of the target quantile. Derman (1957) proves that his biased coin random walk rule, asymptotically, causes the empirical mode to be within Δ of μ for arbitrary Γ . This is also true for BCD (cf., Durham and Flournoy, 1994) and for other biased coin designs constructed using random walk theory (cf.,

Durham and Flournoy, 1995). In the continual reassessment method, one goal of the design is to place design points sequentially closer to the target quantile, leaving the value of the last design point as the estimate of the target quantile. This is similar in spirit to using the mode as an estimator for the random walk rules. The disadvantage of these estimators for the continual reassessment method and the random walk rules is that they can only be taken from a discrete set of values. Other estimators, such as the maximum likelihood estimator, can take on values between assigned dose levels, which is attractive in the event that the target quantile falls below the lowest dose level or above the highest dose level.

Many have studied the use of the empirical mean

$$\bar{X}(n) = \frac{\sum_{i=1}^K x_i N_i(n)}{n}, \tag{5}$$

together with modifications, as an estimator of μ when $\Gamma = 0.50$, including Dixon and Mood (1948), Brownlee, Hodges, and Rosenblatt (1953), Dixon (1965), Wetherill and Chen (1965), Wetherill, Chen, and Vasudeva (1966), Tsutakawa (1967), Hsi (1969), Choi (1971, 1990), and Kershaw (1987). Wetherill (1963) and Cochran and Davis (1964) compared this estimator from Dixon and Mood's up-and-down design with other procedures including the method of stochastic approximation (Robbins and Monro, 1951). These studies relied explicitly on a parametric model of the response function or on simulations.

4.3. A Comparison of Estimators

A simulation study of the performance of the maximum likelihood estimator, the empirical mean, and the mode is given in Table 2 for sample sizes of 25 and 50. The start-up rule described in Section 3 was used with BCD. For small samples, the maximum likelihood estimator is relatively unbiased, but has a fairly large variability. A big disadvantage to maximum likelihood estimation in the logistic model is the existence problems that arise when Silvapulle's (1981) conditions are violated. Nonconvergence is likely to occur from time to time with small samples. At a sample size of 50, the maximum likelihood estimator is much more efficient.

The mode tends to be biased and has fairly large variability for small and medium-sized samples. The empirical mean has the smallest variability of all the estimators at all the quantiles studied here.

5. Redesigning an Experiment in Bone Marrow Transplantation: An Example

Flournoy (1993) described an experiment in which the target proportion of observed toxicities was $\Gamma = 0.10$. Possible dosages of cyclophosphamide (mg/kg/day), $x_0 = \{31, 49, 67, 85, 103, 121\}$, were identified according to the Bayesian method of Tsutakawa (1980), which minimizes the posterior variance around the target dose μ . An up-and-down strategy was conducted on x_0 . First, using the same design points as were used by Flournoy (1993), we show the effect that using the BCD would have had on her experiment. Then we describe an alternative method for selecting design points.

Table 2
Performance of quantile estimators for the random walk rule in Example 1 (8000 replications)

Γ	True μ	$\hat{\mu}(MLE)$			Empirical mean		Empirical mode	
		Bias	S.D.	% convergent	Bias	S.D.	Bias	S.D.
$p = [0.05, 0.10, 0.25, 0.35, 0.50, 0.70], n = 25$								
0.10	2.0	−0.03	1.31	83	0.47	0.68	0.11	0.91
0.20	2.9	−0.05	1.52	91	−0.08	0.69	−0.31	1.00
0.25	3.0	0.34	1.34	94	−0.03	0.67	−0.08	1.02
0.33	3.9	−0.05	1.29	95	−0.63	0.62	−0.51	1.00
0.50	5.0	−0.09	1.04	95	−0.87	0.42	−0.39	0.82
$p = [0.05, 0.10, 0.25, 0.35, 0.50, 0.70], n = 50$								
0.10	2.0	−0.08	0.81	97	0.14	0.48	−0.23	0.70
0.20	2.9	−0.01	0.55	100	−0.15	0.53	−0.39	0.83
0.25	3.0	0.28	0.52	100	0.04	0.52	−0.12	0.86
0.33	3.9	−0.07	0.51	100	−0.39	0.48	−0.40	0.83
0.50	5.0	−0.11	0.63	100	−0.61	0.32	−0.31	0.60

In designing Flournoy's experiment, the response function was assumed to be logistic with parameterization $\text{logit}(p) = \theta + \tau x$. Prior distributions were elicited for θ and τ , from which we obtained the median priors ($\theta_{0.50} = -6.2648$ and $\tau_{0.50} = 0.0547$) and the 97.5th percentiles ($\theta_{0.975} = -4.9805$ and $\tau_{0.975} = 0.0633$), which we call conservative priors. Inserting these priors into the equation $\mu = \{\text{logit}(0.10) - \theta\}/\tau$ yields the median and conservative priors of the target, $\mu_{0.50} = 74.4$ and $\mu_{0.975} = 44.0$, respectively. Note that the dosages in x_0 are centered around $\mu_{0.50}$ with a step size of $\Delta = 18$ between them.

Empirical results after the experiment was concluded indicate that the dose-response function increased sharply, much more sharply than even the conservative prior (see Figure 1). Silvapulle's (1981) conditions were violated, and hence maximum likelihood estimates of the logistic response function did not exist. Here, the data set is augmented by one additional trial, that was assumed to produce a failure, in order to allow maximum likelihood estimation. We call the estimates based on the augmented data set ($n = 34$) empirical estimates of the logistic response function and use these for illustrative purposes. The empirical estimates are $\theta_{\text{emp}} = -4.7$ and $\tau_{\text{emp}} = 0.0974$. The empirical shape parameter is 50% larger than the conservative prior, $\tau_{0.975}$. Furthermore, the empirical response function is shifted downward considerably; the mean of the conservative prior and empirical response functions are $-\theta_{0.975}/\tau_{0.975} = 78.7$ mg/kg and $\theta_{\text{emp}}/\tau_{\text{emp}} = 48.3$ mg/kg, respectively.

The consequences of the conservative prior response function's being so far below the empirical response function were severe. Flournoy's (1993) design called for an *ad hoc* modified up-and-down rule to be used on the design points in x_0 . The rule was to treat in groups of four; increase the dose if no toxicity was seen; use the same dose if there were one out of four toxicities; and decrease the dose if there were two or more toxicities. This was fortunate because it caused only lower dose levels in x_0 to be used. However, the data indicate that the range of dosages in the design failed to cover the target dose; the empirical target dose is $\mu_{\text{emp}} = 25.7$ mg/kg, which is below the lowest dose (31 mg/kg) in the design. Certainly, having more information about the response function at lower dosages would have been useful. In this section, we show that using BCD would have provided more useful information with the same number of patients, even when starting far from the target.

5.1 Allocation Proportions Across the Design Lattice

Figure 2 shows the treatment distribution after 34 trials that can be expected using BCD under the median prior, the conservative prior, and the empirical response functions. Everything except the treatment allocation rule is the same as was used by Flournoy (1993); namely, the design lattice is x_0 and the initial dose is $x_3 = 67$, which is the nearest possible dose below $\mu_{0.50}$.

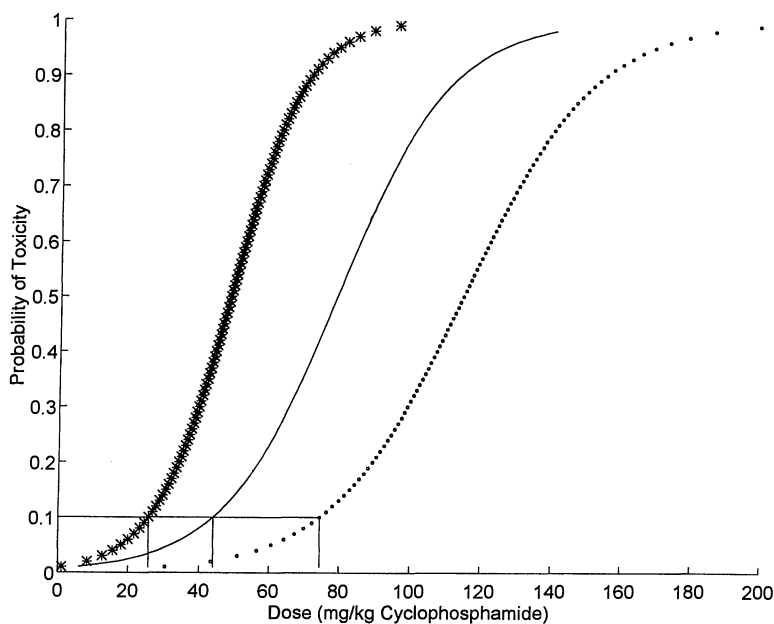


Figure 1. Dose-response functions under three different assumptions with $P\{\text{toxicity} \mid \mu\} = 0.10$ indicated. $\circ \circ \circ$, median prior; —, conservative prior; * * *, empirical.

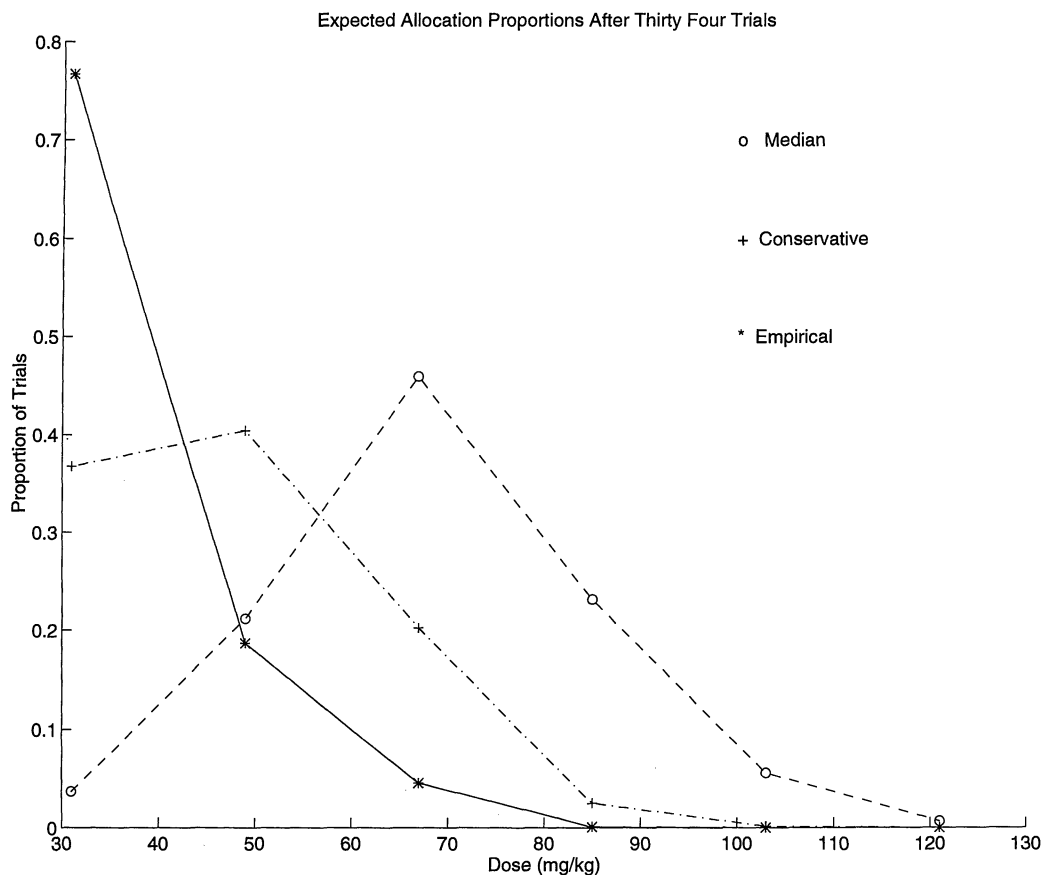


Figure 2. Expected proportions of subjects allocated at each dose level in x_0 after 34 trials under the median, conservative, and empirical response functions.

Figure 3 shows the expected allocation proportions for trials 1 through 34 assuming the empirical response function is operational, but the interval between dosages is reduced from 18 to 6 mg/kg. In this case, the nearest possible dose below $\mu_{0.50}$ is 71.4. Taking the initial dose to be $X(1) = 71.4$, the figure shows how the treatment distribution evolves when the initial dose is distant from μ . We motivate the selection of the reduced spacing in Section 5.5.

For small samples, symmetry in the treatment distribution results from selecting the initial treatment to be close to μ . For large samples, we expect the treatment distribution to become approximately symmetric, even when the initial dosage is far from the target, so long as the design lattice is fine enough to prevent treatments from bunching up at a boundary. Figure 3 shows how using a smaller step size permits more symmetry in the treatment distribution to be maintained as the empirical response function shifts this distribution away from the initial treatment.

5.2 The Empirical Mean as an Estimator of μ

When treatments concentrate symmetrically around the target, any measure of central tendency may be used to estimate μ . In particular, when symmetric treatment distributions are obtained (approximately), the mean of the treatment distribution, given in (5), is a plausible estimate of the target dose.

Figure 4 shows the expected average dosages, $E\{\bar{X}(1)\}, \dots, E\{\bar{X}(34)\}$ with a bias correction, approaching their asymptotes with bounds one standard deviation above and below. These bounds are shown under the median prior, the conservative prior, and the empirical response functions on both the coarse and fine mesh. Note at $n = 1$ the bounds for BCD converge at the initial dosage plus the bias correction. Table 3 shows the expected performance of the average dose when used to estimate μ .

Consider Figure 4 and Table 3 together. Under the median prior, where the initial dose is close to the target, the expected average dose changes only very slightly as trials accumulate. The

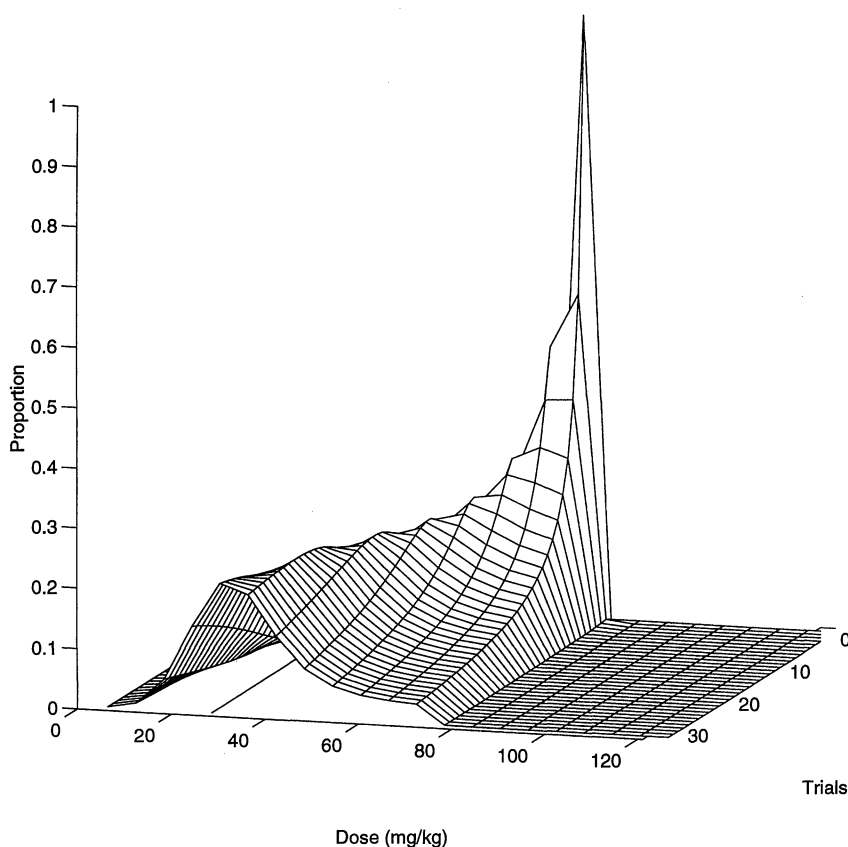


Figure 3. Expected proportions of subjects allocated at each dose level under the empirical response function. We interpolate between the points that occur at $n = 1, \dots, 34$ and each possible dose level in x_1 .

asymptotic average dose with the bias correction (see Section 5.5) is within rounding error of the target dose. For both the conservative prior and empirical response functions, the lack of design points below the target causes the treatment distribution to skew, as shown in Figure 2, and this skewness causes $E\{\bar{X}(n)\}$ to overestimate the target dose, even in the limit. However, when a sufficiently fine mesh is used, as when a step size of 6 is used with the empirical response function, the average dose performs well. The bias correction is seen to improve the performance of this estimator when the allocation distribution is roughly symmetric.

5.3 An Adjusted Empirical Mean Estimator of μ

For large samples or when the starting treatment is close to μ , it is not surprising that the treatment mean underestimates μ by a small amount. Durham and Flournoy (1994) found that, if the response function is logistic and if design points are equally spaced but close enough together that the natural tails of the treatment distribution can evolve without bunching up at a boundary, then the stationary treatment distribution using BCD is a mixture of two discrete normal distributions with means $\mu \pm 0.5\Delta$; that is,

$$\pi_k \approx (1 - \Gamma)\psi\left(\frac{\mu - 0.50\Delta}{\sqrt{\Delta/\tau}}\right) + \Gamma\psi\left(\frac{\mu + 0.50\Delta}{\sqrt{\Delta/\tau}}\right), \quad (6)$$

where

$$\psi(z_k) = \frac{\exp(-z_k^2/2)}{\sum_{j=1}^K \exp(-z_j^2/2)}.$$

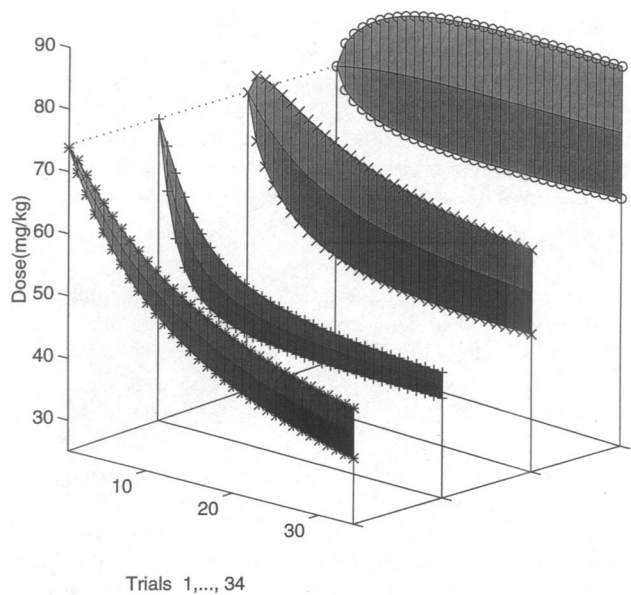


Figure 4. Empirical estimates of the target dose ± 1 standard deviation with bias correction for $n = 1, \dots, 34$ are displayed as a mesh. Four such meshes differ in the underlying response functions and sets of possible dosages. $\circ$ indicates bounds under the median prior response function on x_0 . $\times$ indicates bounds under the conservative prior response function on x_0 . $+$ indicates bounds under the empirical response function on x_0 . $*$ indicates bounds under the empirical response function on x_1 .

Thus, if the underlying response function is the median prior, we expect treatments allocated according to BCD on x_0 to center around the weighted means of the two discrete normal distributions, i.e.,

$$\begin{aligned} x_0 \cdot \pi &\approx (1 - \Gamma)(\mu_{0.50} - \Delta 0.50) + \Gamma(\mu_{0.50} + \Delta 0.50) \\ &= \mu_{0.50} - \Delta(0.50 - \Gamma) = 67.3, \end{aligned} \tag{7}$$

which is very close to the asymptotic mean of the treatment distribution, $x_0 \cdot \pi = 67.5$. The form of (7) indicates that, so long as Δ is positive and $\Gamma \neq 0.50$, (5) will be a biased estimator of μ and that (ignoring for the moment bias induced by selecting an initial treatment far from μ) a better estimator is

Table 3

Using the expected average dose to estimate the quantile corresponding to 10% toxicity								
Step size Δ : Target dose μ : Correction:	Response function							
	Median prior		Conservative prior		Empirical			
	18		18		18		6	
	74.4		44.0		25.7		25.7	
	No	Yes ¹	No	Yes ¹	No	Yes ¹	No	Yes ²
$E\{\bar{X}(n)\},$ $n \rightarrow \infty$	67.5	74.7	42.9	50.1	34.1	41.3	23.4	25.8
$E\{\bar{X}(34)\}$	68.4	75.6	48.9	56.1	36.0	43.2	37.3	39.7
$E\{\bar{X}(1)\}$	67.0	74.2	67.0	74.2	67.0	74.2	71.4	73.8

¹ Bias correction 7.2.
² Bias correction 2.4.

$$\hat{\mu}(n) = x_0 \cdot \{N(n)/n\} + \Delta(0.50 - \Gamma). \quad (8)$$

Note that the bias correction is always less than the step size Δ . The expectation of this estimator is shown in Figure 4 with bounds at ± 1 standard deviation. Note from Table 3 that, when the treatment distribution is symmetric asymptotically, the correction is consistent for μ . Thus when the treatment distribution is evolving symmetrically, but an estimate of μ is desired before $E\{\bar{X}(n)\}$ levels off, the asymptote of (8) could be projected from the series and used as the estimator of μ .

When the treatment distribution exhibits asymmetry, the step size should be reduced if possible. Of course, more robust measures of central tendency could be used as well as the other estimators discussed in Section 4.

5.4. Toxic Events

Using random walk theory (cf., Flournoy et al., 1995), one can calculate the proportion of toxic responses to be expected using BCD under various hypothetical response functions. The expected proportions of subjects having toxicity, $E\{R_1(n)/n\}, \dots, E\{R_K(n)/n\}$, is given by

$$E\{R(n)/n\} = \sum_i E\{R_i(n)/n\} = \sum_i p_i E\{N_i(n)/n\}.$$

Figure 5 displays the proportion of subjects expected to have a toxic response, $E\{R(n)/n\}$, for $n = 1, \dots, 34$, with bounds one standard deviation above and below, approaching their asymptotes. The four scenarios differ in the underlying response functions and sets of possible dosages. They are separated from each other according to the asymptotic average dose that results from these differences. Key statistics from Figure 5 are compared with asymptotic results in Table 4.

Of course at $n = 1$, the expected proportion of toxicities is the probability of toxicity. We focus initially on these points in Figure 5 for reference. As expected, starting just below μ , under the median prior, the probability of toxicity for the first subject [$p = 0.07$ at $X(1) = 67$ on x_0] is just below the target toxicity 0.10. Since the starting dose is higher than μ , the initial probability of toxicity is increased under the conservative prior response function [$p = 0.32$ at $X(1) = 67$ on x_0] and higher yet with both the coarse and fine mesh under the empirical response function [$p = 0.86$ at $X(1) = 67$ on x_0 and $p = 0.90$ at $X(1) = 71.4$ on x_1].

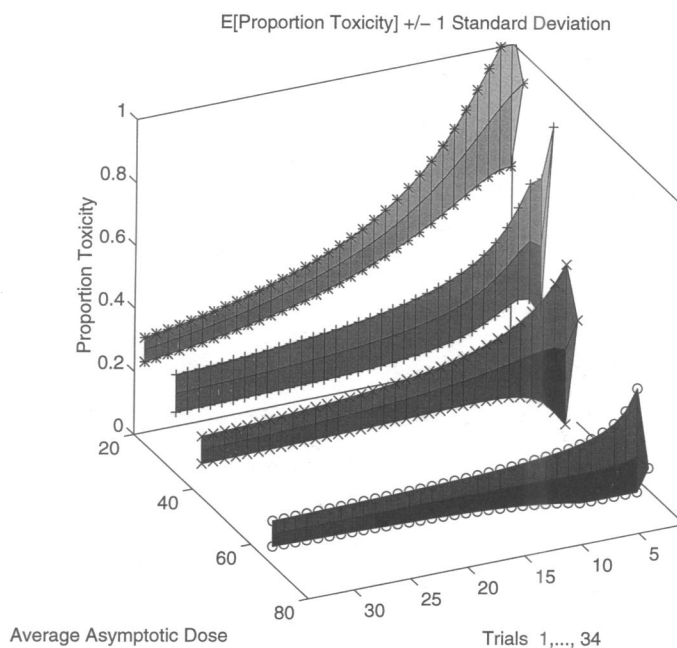


Figure 5. The proportions of subjects expected to have a toxic response to treatment ± 1 standard deviation for $n = 1, \dots, 34$ are displayed as a mesh. Four such meshes differ in the underlying response functions and sets of possible dosages. $\circ$ indicates bounds under the median prior response function on x_0 . $\times$ indicates bounds under the conservative prior response function on x_0 . $+$ indicates bounds under the empirical response functions on x_0 . $*$ indicates bounds under the empirical response functions on x_1 .

Table 4
Expected proportion of toxicities targeting $\Gamma = 0.10$

	Response function			
	Median prior	Conservative prior	Empirical	
Step size Δ	18	18	18	6
Starting dose x_1	67.0	67.0	67.0	71.4
Target dose μ	74.4	44.0	25.7	25.7
$E\{R(n)/n\}, n \rightarrow \infty$	0.10	0.12	0.22	0.10
$E\{R(34)/34\}$	0.10	0.15	0.25	0.30
$E\{R(1)\}$	0.07	0.32	0.86	0.90

Under the median prior, the expected proportion of subjects having a toxic response remains low as trials accumulate [$0.069 \leq E\{R(n)/n\} \leq 0.113, n = 1, \dots, 34$, with asymptote at $\Gamma = 0.10$]. Under the conservative prior response function on x_0 , $E\{R(n)/n\}$ drops immediately and remains below 0.25 for $n \geq 4$. Then $E\{R(n)/n\}$ decreases more slowly [$E\{R(34)/34\} = 0.15$] with asymptote [$\lim E\{R(n)/n\} = 0.12$] held above Γ by the limitations of the restricted design lattice. The restrictions in x_0 are even more problematic under the empirical response function. Although $E\{R(n)/n\}$ drops immediately and remains below 0.50 for $n \geq 4$ and below 0.30 for $n \geq 15$, $E\{R(34)/34\} = 0.25$ and $\lim E\{R(n)/n\} = 0.22$. Under the empirical response function on x_1 , $E\{R(n)/n\}$ drops rapidly in spite of the step sizes being one-third the size as on x_0 : $E\{R(n)/n\}$ remains below 0.50 for $n \geq 13$ and below 0.30 for $n \geq 34$ with asymptote at $\Gamma = 0.10$.

Comparing the expectations produced under the median prior response function on x_0 with those produced under the empirical response function on x_1 demonstrates that, asymptotically, if the design lattice is sufficiently fine to permit the treatment distributions to be reasonably symmetric about the target, the BCD will cause the total proportion of toxic responses to equal the target under any monotone increasing response function.

With the ability to perform such calculations, estimates of the underlying response function as a study progresses can be used to decide whether the step size should be increased or decreased to achieve the objectives of the study. Of course, such adjustments may raise interesting issues of interim analysis. In fact, if $\Gamma = 0.5$ and the dosages are equally spaced but the interval Δ is permitted to shrink with each trial as $\Delta(n) = \Delta/n$, then the up-and-down method becomes stochastic approximation (Robbins and Monro, 1951). Levitt (1971, p. 471) notes that

... Although the method of reducing the step size after every trial has distinct theoretical advantages, its practical implementation may require fairly complicated equipment. A simple approximation is to reduce the step size by a proportionate amount after one or more complete runs (a run is defined to be a series of steps in one direction only). A convenient practical procedure is to halve the step size after the first, third, seventh, ..., 2^{n-1} th run.

Empirical sampling trials (Wetherill, 1963) indicate that, if the spread and location of the response function are known within reasonable limits ..., then a very good approximation to the Robbins and Monro efficiency is obtained by halving the step size only once, preferably after the 3rd run.

We now motivate the design lattice x_1 .

5.5 Design to Control Toxic Events

Another goal is to avoid highly toxic dose levels. Suppose we want 97.5% of the treatments to be less than or equal to some unknown dose μ_1 for which $P\{Y=1 \mid \mu_1\} = \Gamma_1, \Gamma_1 > \Gamma = 0.10$. From (6), it can be seen that the stationary allocation distribution for the BCD is a mixture of two discrete normal distributions with common scale parameter Δ/τ ; thus, the spread around μ can be controlled by setting $2(\Delta/\tau)^{1/2}$ equal to $\mu_1 - \mu$ and solving for Δ to yield

$$\Delta = \frac{\{\text{logit}(\Gamma_1) - \text{logit}(\Gamma)\}^2}{4\tau}. \tag{9}$$

A prior estimate τ can be used to determine Δ . One could consider updating Δ at each trial by sequentially estimating τ or updating in stages when the empirical and prior estimates differ substantially.

Consider Δ to be fixed for the duration of the experiment. The step size found by replacing τ in equation (9) with $\tau_{0.50}$ for several degrees of restriction is

Γ_1	0.50	0.40	0.30	0.20
Δ	7.72	6.00	3.96	1.68

If we wish to protect against treating at levels for which $P\{toxicity\} \geq 0.40$, we set $\Delta = 6$ and take the lattice of design points to be $\mu_{0.50} \pm 3, \mu_{0.50} \pm 9 \dots$. For Flournoy’s design, such a lattice would be $x_1 = \{5.4, \dots, 71.4, 77.4, \dots\}$. Using this lattice of points with the BCD, the empirical response function would allocate treatments as shown in Figure 3 for experiments of size $n = 1, \dots, 34$. The effects of using this step size on the treatment mean and on expected toxicities were described in Sections 5.2–5.4 and are shown in Tables 3 and 4 and in Figures 4 and 5.

In this section, we have demonstrated how exact distribution theory is very useful in planning studies. In particular, being able to calculate the expected proportion of toxic events for experiments of any sample size and any underlying response function is an exciting development in the design of phase I clinical trials.

6. Discussion

We reiterate the advantages of random walk rules. They satisfy an ethical imperative of minimizing unnecessary assignment of patients to toxic dose levels while allowing for targeting of a specific quantile. Trials tend to concentrate in a region about the target quantile, even when there is a large number of dose levels. Random walk rules do not rely on any distributional assumption for the underlying dose–response curve. They generate designs that are comparable to those of other methods, such as the continual reassessment method, yet they are simpler to implement (for example, the continual reassessment method requires numerical integration). Furthermore, the entire distribution theory for the random walk rules (both finite and asymptotic) has been worked out. No other sequential designs for quantal response of which we are aware have a completely workable distribution theory. It is important to note that many *ad hoc* up-and-down procedures have been used (e.g., Storer, 1989; Flournoy, 1993; the standard method of Korn et al., 1994) without theoretical analysis of their consequences.

It is commonly accepted that phase I clinical trials will focus on toxicity rather than efficacy. However, beyond this common feature, the term is applied in a multitude of experimental settings. Distinguishing features include the experimental objectives and the amount of historical data available on toxicity. Phase I clinical trials include: (1) experiments where the goal is to screen a potentially active agent out of a very large number of potential agents, with a drug passing the screen subjected to further study; (2) experiments where the goal is to investigate new combinations of research drugs that have a long history of use individually or in other combinations; and (3) experiments where drugs with a long history of use for some diseases are tested in patients with other diseases or other stages of the same disease.

In the first situation, the sheer volume of drugs to be screened calls for methods that are quick and often crude, with protection being built into the stages that a new drug must pass through to come to market. On the contrary, when a large amount of historical data on toxicity exists, phase I trial objectives may be much more specific and the objective may be to obtain a regimen that achieves a relatively small improvement (say 20%) over a control. The new regimen may even contain the same drugs as the control, but given according to an alternate schedule. In such cases, it is important that the new regimen be no more toxic than the control. Thus, toxicity in such trials must be determined more precisely than in drug screening studies.

Korn et al. (1994) describe a method as standard for phase I clinical trials. This *ad hoc* method was developed for screening drugs, and it permits highly toxic agents to be identified quickly and allows other drugs to pass to a further stage of testing. It is invariant with respect to levels of toxicity that may be deemed tolerable. Simulation evidence suggests it targets approximately the 20th or 25th percentile.

We have assumed throughout this paper that the lattice and mesh of design points is prespecified. However, our review of Flournoy (1993) clearly demonstrates the undesirable consequences of having a prior response function far from the empirical response function. Hence, the set of dose levels selected as part of the design should be reviewed and revised as needed as the study progresses.

While, for maximum likelihood purposes, we have assumed a logistic model, it should be clear that any location-scale family could be selected for parametric analysis and the performance of

the random walk rules themselves are independent of that choice. This is especially important in Bayesian procedures, where the allocation rule depends on the underlying distributional assumption of a parametric prior distribution. The effects of model misspecification in these parametric procedures merits further investigation.

Random walk rules (and most other designs for dose-response studies) are applicable when patient response is obtainable fairly quickly. Fortunately, in many studies, information on patient response is obtainable quickly. When it is not, the one patient per cohort model may be impractical. Random walk rules can be extended to apply to groups of patients in each cohort, provided a rigid definition of toxicity is maintained. Grouped random walk rules were first proposed by Anderson et al. (1946) for estimating the median. Further theoretical work was done by Tsutakawa (1967), and simulations targeting extreme percentiles can be found in Hsi (1969) and Wetherill and Glazebrook (1986). Exact properties of grouped random walk designs are forthcoming (H. Wu, unpublished data).

We have also discussed only fixed-sample designs. The full benefit of adaptive designs, such as this random walk rule, is more likely to be attained by incorporating stopping rules. Combining random walk rules with the concepts of sequential analysis opens an array of potential theoretical problems that should be well worth the effort.

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RÉSUMÉ

Nous décrivons une famille de règles permettant de régir des marches aléatoires au moyen desquelles on peut ensuite affecter, de façon séquentielle, des niveaux de doses aux patients d'une étude de dose-réponse ou d'un essai clinique de phase I. C'est en se basant sur la dose du patient qui l'a précédé que l'on affecte à chaque nouveau patient la dose immédiatement supérieure, la même dose ou la dose immédiatement inférieure, en fonction de probabilités déterminées à la fois par des considérations éthiques et par le résultat obtenu sur le patient précédent. Nous montrons qu'il est alors possible de choisir ces probabilités de manière à ce que les niveaux de dose effectivement attribués soient centrés, de façon unimodale, autour d'un quantile prédéterminé. Nous abordons ensuite le problème de l'estimation dudit quantile en calculant, sous l'hypothèse d'une distribution logistique à deux paramètres, l'estimateur du maximum de vraisemblance—ainsi que sa variance—, que nous comparons avec des estimateurs non paramétriques. Les méthodes de marche aléatoire présentent réellement des avantages: elles sont faciles à mettre en oeuvre, et la théorie permet de connaître précisément les distributions attendues, aussi bien finies qu'asymptotiques. En guise d'illustration, nous calculons les propriétés finies et asymptotiques d'une marche aléatoire particulière et donnons des exemples de son utilisation. À l'évidence, disposer d'une théorie sur les distributions finies et pouvoir l'appliquer permet d'établir les propriétés du plan choisi sans avoir à recourir aux simulations. Dans le cas de petits échantillons, la détermination exacte—issue de la théorie—des propriétés de la marche aléatoire précédemment évoquée permet de les comparer à leur avantage aux propriétés de la méthode proposée par O'Quigley (*continual reassessment method*) évaluées par simulation.

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APPENDIX

Derivation of Asymptotic Variance Approximation of $\hat{\mu}$ Under a Logistic Model

If p_i is of the form in (3), then $\partial p_i / \partial \alpha = -\beta^{-1} p_i q_i$ and $\partial p_i / \partial \beta = -\beta^{-1} \{(x_i - \alpha) / \beta\} p_i q_i$. Then we can use the chain rule to find

$$\begin{aligned} \frac{\partial^2 \ell}{\partial \alpha^2} &= \frac{\partial}{\partial \alpha} \left\{ \sum_{i=1}^K \frac{\partial \ell}{\partial p_i} \frac{\partial p_i}{\partial \alpha} \right\} = -\frac{1}{\beta^2} \sum_{i=1}^K N_i(n) p_i q_i, \\ \frac{\partial^2 \ell}{\partial \alpha \partial \beta} &= -\frac{1}{\beta^2} \sum_{i=1}^K \left(\frac{x_i - \alpha}{\beta} \right) N_i(n) p_i q_i + \frac{1}{\beta^2} \sum_{i=1}^K [R_i(n) - p_i N_i(n)], \\ \frac{\partial^2 \ell}{\partial \beta^2} &= -\frac{1}{\beta^2} \sum_{i=1}^K \left(\frac{x_i - \alpha}{\beta} \right)^2 N_i(n) p_i q_i + \frac{2}{\beta^2} \sum_{i=1}^K \left(\frac{x_i - \alpha}{\beta} \right) [R_i(n) - p_i N_i(n)]. \end{aligned}$$

Then the Fisher's information is

$$I \equiv -E \begin{bmatrix} \frac{\partial^2 \ell}{\partial \alpha^2} & \frac{\partial^2 \ell}{\partial \alpha \partial \beta} \\ \frac{\partial^2 \ell}{\partial \alpha \partial \beta} & \frac{\partial^2 \ell}{\partial \beta^2} \end{bmatrix}.$$

Because the first derivative of the log-likelihood is always a zero-mean martingale, we have that

$$E\{R_i(n) - p_i N_i(n)\} = 0.$$

Hence,

$$I = \frac{1}{\beta^2} \sum_{i=1}^K \begin{bmatrix} 1 & \frac{(x_i - \alpha)}{\beta} \\ \frac{(x_i - \alpha)}{\beta} & \left(\frac{x_i - \alpha}{\beta} \right)^2 \end{bmatrix} E[N_i(n)] p_i q_i,$$

and the inverse of the Fisher's information is

$$\mathbf{I}^{-1} = \frac{\frac{1}{\beta^2} \sum_{i=1}^K \begin{bmatrix} \left(\frac{x_i - \alpha}{\beta}\right)^2 & -\frac{(x_i - \alpha)}{\beta} \\ -\frac{(x_i - \alpha)}{\beta} & 1 \end{bmatrix} \mathbf{E}[N_i(n)]p_i q_i}{|\mathbf{I}|},$$

where

$$\begin{aligned} |\mathbf{I}| &\equiv \frac{1}{\beta^4} \left\{ \sum_{i=1}^K \left(\frac{x_i - \alpha}{\beta} \right)^2 \mathbf{E}[N_i(n)]p_i q_i \right\} \left\{ \sum_{i=1}^K \mathbf{E}[N_i(n)]p_i q_i \right\} \\ &\quad - \frac{1}{\beta^4} \left\{ \sum_{i=1}^K \left(\frac{x_i - \alpha}{\beta} \right) \mathbf{E}[N_i(n)]p_i q_i \right\}^2 \\ &= \frac{1}{\beta^6} \sum_{i < j} \left\{ \mathbf{E}[N_i(n)]\mathbf{E}[N_j(n)]p_i q_i p_j q_j (x_i - x_j)^2 \right\}. \end{aligned}$$

This, then, defines the variance-covariance matrix for $(\hat{\alpha}, \hat{\beta})$. To find the variance of $\hat{\mu}$, we use (2) to obtain $\text{var}(\hat{\mu}) = \text{var}(\hat{\alpha}) + [F^{-1}(\Gamma)]^2 \text{var}(\hat{\beta}) + 2[F^{-1}(\Gamma)]\text{cov}(\hat{\alpha}, \hat{\beta})$, recalling that, for the logistic distribution, $F^{-1}(\Gamma) = \ln(\Gamma/(1 - \Gamma))$. As $\hat{p}_i$ is consistent for p_i and $\hat{\alpha}$ and $\hat{\beta}$ are consistent for α and β (Rosenberger, Flournoy, and Durham, 1997), respectively, we obtain the natural estimator of $\text{var}(\hat{\mu})$ given in (4) by replacing $\mathbf{E}[N_i(n)]$ with $N_i(n)$, p_i with $\hat{p}_i$, q_i with $\hat{q}_i$, α with $\hat{\alpha}$, and β with $\hat{\beta}$. Alternatively, we could replace $\mathbf{E}[N_i(n)]$ with the asymptotic mean $n\pi_i$.